18Gbps, 50mW Reconfigurable Multi-mode SHA Hashing Accelerator in 45nm CMOS

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***Abstract*—A multi-mode Secure Hashing Algorithm (SHA) accelerator is fabricated in 45nm CMOS and occupies 0.0625mm^2 with 18Gbps throughput and total power consumption of 50mW. The reconfigurable hardware accelerator computes SHA-1/224/256/384/512 message-digest using unified SHA bit-slices and configurable compression circuits resulting in 40% area reduction and <3% performance overhead for reconfiguration with 23Gbps peak throughput in SHA-224/256 modes. SHA frequency ranges from 21MHz-1.8GHz across 320mV-1.35V supply voltage range.**

I. Introduction

Hash functions are important cryptographic primitives used to generate digital fingerprints or signatures [1] in public-key encryption workloads and internet protocols, such as Internet Protocol Security (IPSEC), Message Authentication Codes (MAC) and Secure Sockets Layer (SSL) [2]. The Secure Hashing Algorithms (SHA) family of hashes provides a variety of digest sizes (160b-512b) with varying number of rounds for a variety of applications. While SHA-1 is the most popular hash in use today, the SHA-2 family of hashes (SHA-224/256/384/512) offers enhanced security in the form of higher pre-image and collision resistance at the cost of higher computational complexity. SHA workloads are among the most performance/power-critical workloads due to the iterative nature of hash computation and the high computational complexity of mapping the parallel 64b additions, bitwise rotations and one-way compression functions on a general-purpose microprocessor execution core.

Several hardware techniques have been used to improve SHA performance [4-9]. These include the use of parallel counters and carry save adders (CSA) [4-6], loop unrolling to mitigate the serial dependence of hash computation [8], delay balancing [5], embedded memories [9] and pipelining [5] to improve throughput. These techniques require significant additional hardware, resulting in higher area and reduced energy-efficiency. The message digest computation of the five SHA algorithms (SHA-1/224/256/384/512) is very similar and can be implemented using common hardware, resulting in significant area savings over a separate datapath approach. This work presents a SHA accelerator based on a unified message-digest bit-slice with minimal performance overhead for configuration. This paper is organized as follows: Section II provides an overview of the Secure Hashing Algorithm.

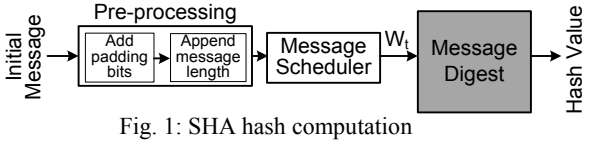


Fig. 1: SHA hash computation

Section III presents the unified SHA bit-slice that can be reconfigured to compute the message digest in five SHA modes of operation. Section IV describes the design of the SHA accelerator. Finally, Section V presents 45nm CMOS measurement results and Section VI concludes the paper.

II. Secure Hash Computation

Hash functions compress messages of arbitrary lengths (<2^{64} bits for SHA-1/224/256 and <2^{128} bits for SHA-384/512) into a fixed-length 160b-512b hash (Fig. 1). During pre-processing, the input messages are split into m-bit blocks (m=512bits for SHA-1/224/256 and m=1024 for SHA-384/512). The last block (with length < m) is appended with "1" followed by series of '0's, with the message length appended to the end of the block. These m-bit blocks are then divided into sixteen 32b/64b words and sent to the scheduler, which prepares the message schedule for hash generation. Message digest computation, the most computationally intensive stage of hash creation, iteratively combines the message schedule (W_t) with a running hash value, generating the final hash when all blocks of the message have been processed. Hash computation begins with a 160b/256b/512b initial hash value stored in the hash register. The multi-mode SHA-1/224/256/384/512 message digest accelerator (Fig. 2) combines W_t from the scheduler FIFO with the round constant (K_t) to generate a new hash which overwrites the value stored

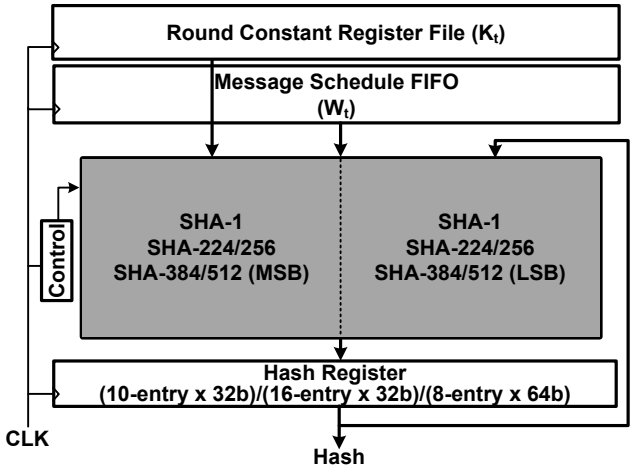


Fig. 2: Multi-mode SHA accelerator organization

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in the hash register [1]. This work describes the implementation of a reconfigurable SHA accelerator that maps the various modes of SHA onto a common datapath by taking advantage of similarities in the operations to enable a multi-mode design with <3% performance overhead for reconfiguration.

The SHA-1 message digest datapath is organized as bitslices operating on 32b operands, with hash values stored in registers A-E (Fig. 3). The critical 'A'-computation involves a 5b left-rotate, an iteration-dependent compression function f_t and five-way 32b addition. These computations iterate for 80 rounds after which A-E values are concatenated to obtain the 160b hash.

$$f_t = \begin{cases} \text{Ch(B,C,D)} = \text{BC} \oplus \bar{B}\text{D}, & 0 \le t \le 19 \\ \text{Parity(B,C,D)} = \text{B} \oplus \text{C} \oplus \text{D}, & 20 \le t \le 39 \\ \text{Maj(B,C,D)} = \text{BC} \oplus \text{BD} \oplus \text{CD}, & 40 \le t \le 59 \\ \text{Parity(B,C,D)} = \text{B} \oplus \text{C} \oplus \text{D}, & 60 \le t \le 79 \end{cases}$$

Message digest computation for the SHA-2 family of hashes uses 32 datapath slices for SHA-224/256 and 64 datapath slices for SHA-384/512 (Fig. 4). Hash values are stored in registers A-H with 64 and 80 iterations for SHA-224/256 and SHA-384/512 respectively. SHA-2 hashes use more complex diffused rotations $\sum_0$ and $\sum_1$ compared to the left rotates used in SHA-1:

$$\text{SHA-224/256: } \Sigma_0(x) = \text{ROTR}^2 \oplus \text{ROTR}^{13} \oplus \text{ROTR}^{22}$$

$$\text{SHA-384/512: } \Sigma_0(x) = \text{ROTR}^{28} \oplus \text{ROTR}^{34} \oplus \text{ROTR}^{39}$$

$$\text{SHA-224/256: } \Sigma_1(x) = \text{ROTR}^6 \oplus \text{ROTR}^{11} \oplus \text{ROTR}^{25}$$

$$\text{SHA-384/512: } \Sigma_1(x) = \text{ROTR}^{14} \oplus \text{ROTR}^{18} \oplus \text{ROTR}^{41}$$

The critical 'A' computation involves 7-way addition, 3-way rotated diffusion and 3-input compression functions. The similarities of operations in the various SHA modes of operation motivate a unified datapath bitslice that can be configured to operate in any one of the 5 modes. The SHA bitslice critical path is further optimized by computing the message digest in redundant carry-save format, thereby postponing the expensive 32b/64b carry-propagation to the end of the cycle, prior to A and E register updates [4-6].

III. UNIFIED SHA BIT-SLICE

A unified SHA datapath is created by mapping the SHA-1 bitslice onto the SHA-2 datapath circuit (Fig. 5). To maximize the overlap between the two bitslices, SHA-1 hash registers A, B, C, D, E are mapped onto SHA-2 registers E, D, F, G and H. The critical 'A' computation of SHA-1 is mapped onto the 'E' circuits of SHA-2. The 30b left rotate required for 'C' generation is implemented using a 2b right rotate. Unused SHA-2 circuits are gated-off during SHA-1 mode of operation (shaded grey in Fig. 5). The SHA bit-slice is optimized to minimize the impact of reconfiguration on SHA-2 message digest computation. Four stages of 3:2 compressor circuits add K_t, W_t, h, Ch, $\sum_1$, d, $\sum_0$ and Maj without propagating carries, producing the outputs S_a and C_a in carry-save representation (Fig. 6). Circuits for 'E' computation are grouped together to perform a 6-way carry-save addition producing outputs S_e and C_e. These circuits are also used in the SHA-1 mode to compute the 'A' output. Thus SHA-1 circuits are incorporated into the SHA bitslice with <3% increase in SHA-2 critical path delay. 2:1 multiplexers select the appropriate inputs for f_t, $\sum_1$ and $\sum_0$ for all five modes of operation. The f_t compression circuit is reconfigured to generate BC⊕(~B)D in SHA-2 modes and for iterations 0-19 in SHA-1 mode (Ch=1, Parity=Maj=0), B⊕C⊕D (when Parity=1, Ch=Maj=0) during iterations 20-39 and 60-79 in SHA-1 mode and BC⊕BD⊕CD (when Maj=1, Ch=Parity=0) during iterations 40-59 in SHA-1 mode of operation (Fig. 7).

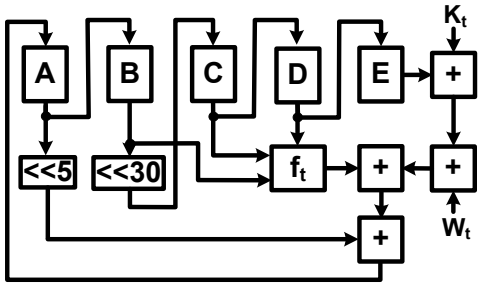


Fig 3: SHA-1 message digest datapath

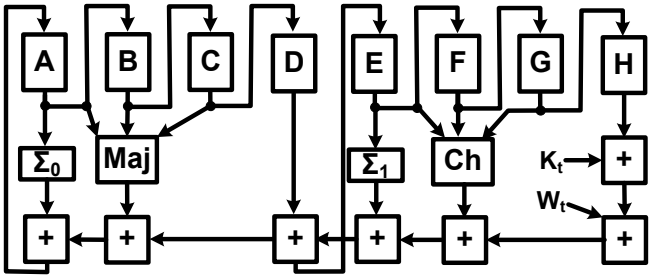


Fig 4: SHA-2 message digest datapath

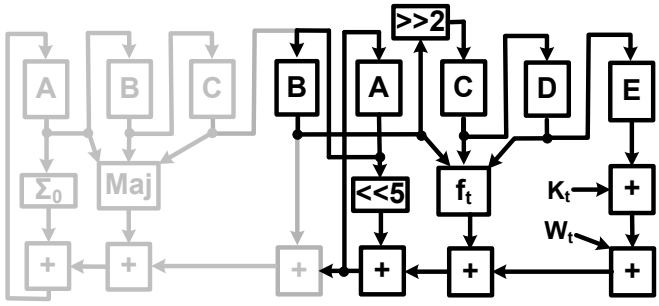


Fig 5: SHA-1 mapped onto SHA-2 datapath

IV. MULI-MODE SHA ACCELERATOR

The multi-mode SHA accelerator is organized as an array of 64 unified SHA bitslices with carry-kill circuits at the 32b boundary (Fig. 8). A pair of 64b adders convert the 'A' and 'E' outputs from carry-save representation to regular binary format. The accelerator can be configured to operate as

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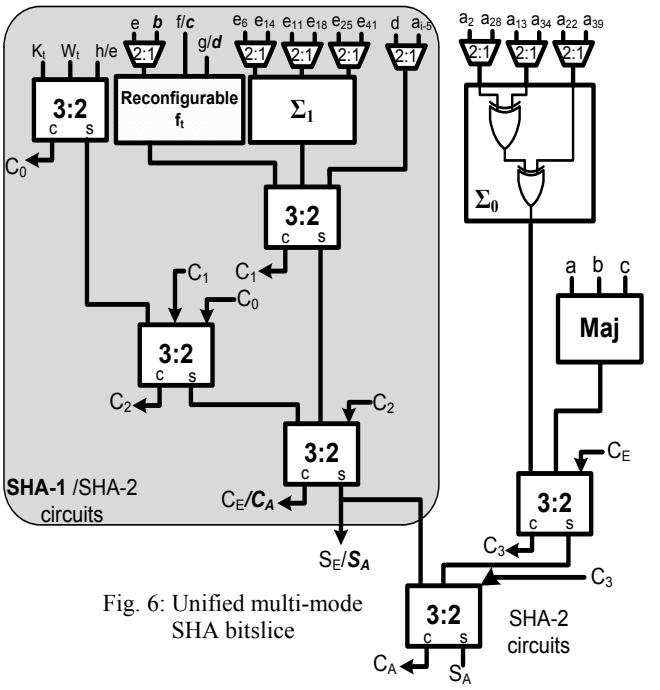


Fig. 6: Unified multi-mode SHA bitslice

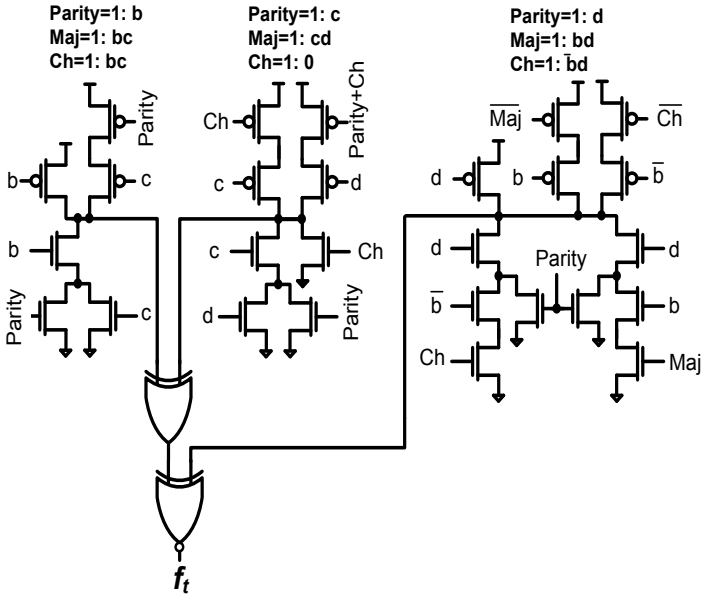
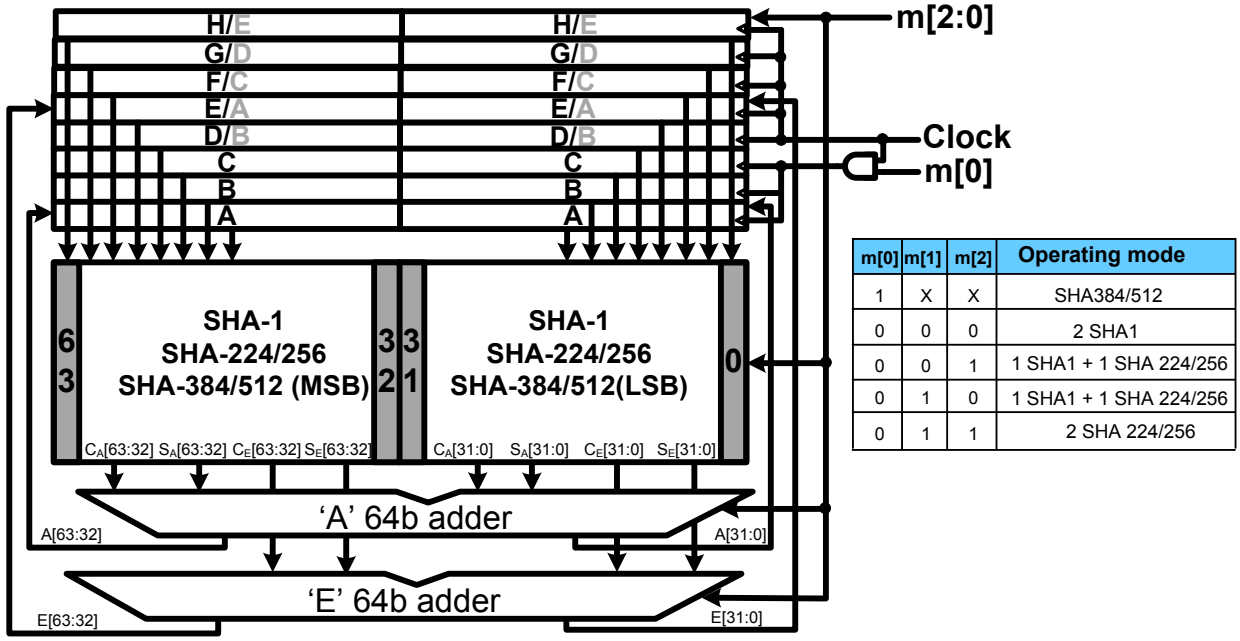


Fig. 7: Reconfigurable f_t compression circuit

parallel SHA-1/224/256 or a single SHA-384/512 message-digest core, with hash values stored in an 8-entry 64b register-file. Three configuration bits are used to reconfigure the accelerator into 1 of 5 modes (i) SHA-384/512 (ii) two parallel SHA-1 (iii & iv) parallel SHA-1 and SHA-224/256 modes (v) two parallel SHA-224/256. In the 32b modes (SHA-1, SHA-224 and SHA-256), carry-kill circuits at the 32b boundary allow parallel operations to compute on either half of the datapath, thus providing 2x higher throughput in these modes. During 32b modes, the hash register file is also reconfigured to operate in parallel 5-entry 32b format for SHA-1 and 16-entry 32b format for SHA-224/256. Unused sections of the register file are clock-gated to reduce unwanted switching activity (Fig. 8). Every cycle, the accelerator processes a new message schedule W_t, completes one iteration of message digest computation, and overwrites the hash register file with the new hash value at the end of the cycle.



m[0]	m[1]	m[2]	Operating mode
1	X	X	SHA384/512
0	0	0	2 SHA1
0	0	1	1 SHA1 + 1 SHA 224/256
0	1	0	1 SHA1 + 1 SHA 224/256
0	1	1	2 SHA 224/256

Fig. 8: Reconfigurable SHA message-digest accelerator

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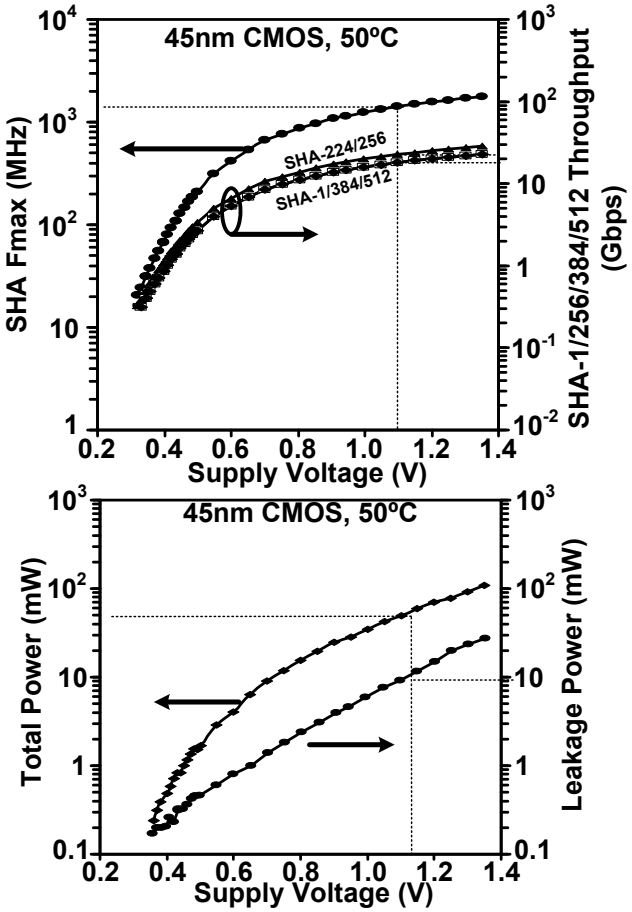


Fig. 9: Frequency, throughput and power measurements in 45nm CMOS

V. 45NM CMOS MEASUREMENT RESULTS

In a 1.1V, 45nm CMOS process [10], the reconfigurable multi-mode SHA accelerator operates at F_{max} of 1.4GHz, with total power consumption of 50mW, measured at 50°C (Fig. 9). Leakage component of total power is 9mW (18% of total power). In SHA-1 mode of operation, the accelerator simultaneously processes two 512b messages producing two 160b hashes with 80 cycle latency, resulting in SHA-1 throughput of 18Gbps at 1.1V, 50°C. Peak throughput of 23Gbps is achieved in SHA-224 and SHA-256 modes of operation at 1.1V, 50°C. Accelerator throughput in SHA-384 and SHA-512 modes is 18Gbps. SHA accelerator F_{max} ranges from 21MHz-1.8GHz, measured across a wide supply voltage range of 320mV-1.35V. The unified SHA datapath results in 40% area reduction compared to a separate datapath implementation, resulting in a compact layout area of 0.0625mm^2 (Fig. 10).

VI. CONCLUSIONS

A reconfigurable multi-mode SHA accelerator targeted for on-die acceleration of SHA message-digest computation in high-performance microprocessors is fabricated in 45nm CMOS technology. The SHA accelerator operates at 1.4GHz with measured throughput of 18-23Gbps in SHA-1/224/256/384/512 modes. A unified SHA bitslice with reconfigurable compression circuits result in 6x higher throughput than previously reported measurements [2], with <3% performance overhead for reconfiguration.

Table 1: 45nm CMOS Measurement summary

Mode	Parallelism	Throughput (Gbps)
SHA-1	2	18
SHA-224	2	23
SHA-256	2	23
SHA-384	1	18
SHA-512	1	18

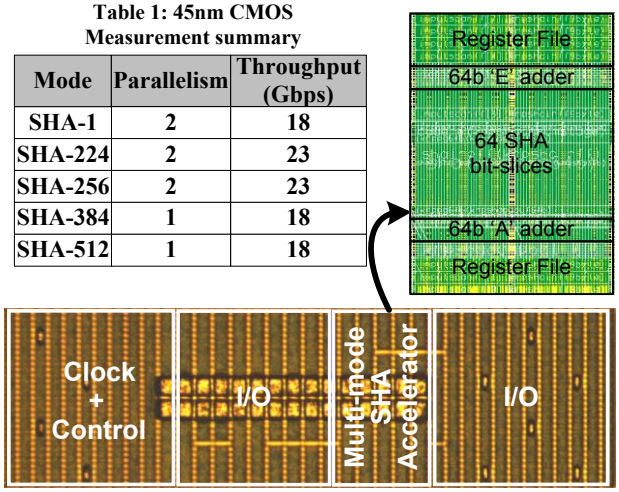


Fig. 10: 45nm die microphotograph and SHA accelerator layout

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